

AI-POWERED INTELLIGENT SALES FORECASTING PLATFORM USING PREDICTIVE ANALYTICS GROUP-1

Team Members & Roles:

No.	Team Member	Role
1	Ritika Kushwaha	Project Lead / Scrum Master, Backend Developer
2	Aditi Bhosure	Database / DevOps Engineer
3	Samraiyam Nallagorla	AI/ML Engineer
4	Tejashwini M	Frontend Developer
5	Rushikesh Arepalli	QA / Test Engineer

Project Overview:

SalesGenie AI is an AI-based sales assistant that brings lead management, AI-based lead analysis, outreach, scoring, recommendations, CRM integration, and sales analytics together in one platform. The project supports sales teams by reducing manual work and presenting useful information for follow-up decisions.

Project Objective:

The main objective of SalesGenie AI is to create an intelligent sales assistant that helps sales teams manage leads in a simpler, faster, and more organized way. The platform is designed to help teams:

- Manage and organize lead and prospect information.
- Analyze lead details and create personalized outreach.
- Prioritize leads with scores and follow-up recommendations.
- Connect CRM and conversation information with the sales workflow.
- View sales performance, pipeline status, and key metrics in one dashboard.

Project Modules:

The SalesGenie AI project is divided into six connected modules that support the sales process from lead management to analytics.

Module 1 - Lead Management & Prospect Database

Manages lead and prospect information.

Module 2 - Lead Intelligence & Company Analysis

Provides AI-supported lead and company insights.

Module 3 - AI Outreach Generation

Creates personalized outreach content.

Module 4 - Lead Scoring & Recommendation Engine

Provides lead scores and next-action recommendations.

Module 5 - Conversation Intelligence & CRM Integration

Connects CRM workflows and provides conversation summaries.

Module 6 - Dashboard & Sales Analytics

Displays conversion, pipeline, KPI, and follow-up information in a central dashboard.

Milestone 4 Delivery Summary:

Milestone	Duration	Development Period	QA / Review	Focus	Acceptance Focus	Status
Milestone 4	Day 26-30	19-Aug-2026 to 21-Aug-2026	24-Aug-2026	Module 6 + Full Integration	Operational dashboard; functional end-to-end workflow; response target <5 sec; UAT target >=85%	Signed Off

Milestone - 4: QA Report

Testing Overview:

Milestone 4 was the final major testing checkpoint of the SalesGenie AI project. The focus was to validate the Dashboard & Sales Analytics module together with the complete integrated application before deployment and project closure.

The milestone covered the analytics API for conversion rate, pipeline value, and sales-cycle KPIs; the Streamlit dashboard and Kanban-style pipeline view; the automated follow-up recommendation panel; end-to-end integration and performance tuning; full regression testing; UAT with sample users; deployment; documentation; and the final demo.

QA activities included functional testing, analytics validation, dashboard and pipeline validation, full regression testing, integration testing, performance review against the <5 second response target, UAT review against the >=85% satisfaction target, defect retesting, and final sign-off.

Testing Objectives:

The main goal of Milestone 4 testing was to confirm that the complete SalesGenie AI platform was stable, usable, and ready for deployment. Testing focused on the following areas:

1. To verify that analytics API results for conversion rate, pipeline value, and KPIs were available to the dashboard.
2. To check the dashboard UI, pipeline Kanban view, and automated follow-up recommendation panel.
3. To validate the complete end-to-end workflow across all six modules.
4. To run full regression testing and confirm that earlier functions continued to work after final integration.
5. To review application response behavior against the <5 second target.
6. To conduct UAT with sample users and capture feedback against the >=85% satisfaction target.
7. To verify that identified issues were resolved before deployment and closure.

Testing Scope:

The testing scope for Milestone 4 was divided into the following areas.

Dashboard & Sales Analytics

- Checking conversion rate, pipeline value, and sales-cycle KPI availability.
- Validating analytics API responses and dashboard display.
- Checking dashboard filters, cards, and metric presentation.

Pipeline and Follow-up Recommendations

- Checking the Kanban-style pipeline view.
- Checking automated follow-up recommendation visibility.
- Reviewing whether recommended next actions appeared with the relevant lead or pipeline information.

Full Integration and Performance

- Testing end-to-end data flow across all six modules.
- Reviewing response behavior against the <5 second target.
- Running full regression testing after integration changes.

UAT, Deployment & Closure

- Conducting UAT with sample users.
- Reviewing satisfaction feedback against the >=85% target.
- Validating the deployed build, final demo, and documentation handover.

Testing Environment & Tools:

Area	Tool / Technology
API Testing	Postman
Backend	FastAPI
Database	PostgreSQL
ORM	SQLAlchemy
Frontend	Streamlit
AI Integration	Gemini / AI API
Automated Testing	PyTest
Deployment	Docker
Version Control	Git
Manual Testing / UAT	Application UI / Browser

Testing Approach:

Milestone 4 used functional, regression, integration, performance, and UAT testing. Dashboard values and pipeline information were checked through the UI and backend workflow. The final integrated application was tested end to end before deployment and project closure.

Test Cases:

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-29	Analytics	Retrieve conversion, pipeline, and KPI data	Analytics API should return required values	Analytics data was available to the dashboard	Pass
TC-30	Dashboard	Display analytics cards and dashboard information	Dashboard should display returned metrics correctly	Dashboard display was validated	Pass
TC-31	Pipeline	View Kanban-style sales pipeline	Pipeline stages should be visible and organized	Pipeline view was validated	Pass
TC-32	Follow-up	Display follow-up recommendations	Recommendations should be shown for relevant workflow	Follow-up panel was validated	Pass

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-33	Integration	Run workflow across all six modules	Modules should communicate correctly end to end	Integrated workflow was validated	Pass
TC-34	Performance	Review application response behavior	Response should meet the <5 second target	Response behavior was reviewed after tuning	Pass
TC-35	Regression	Run full regression test suite	Existing features should continue to work	Regression testing was completed	Pass
TC-36	UAT	Conduct UAT with sample users	User feedback should meet >=85% satisfaction target	UAT review was completed	Pass
TC-37	Deployment	Validate deployed build and final demo flow	Deployed application should function correctly	Deployment validation was completed	Pass

Test Execution Metrics:

Milestone 4 testing covered the documented final-module, integration, regression, UAT, and deployment scenarios. Two issues identified during initial testing were resolved and retested before final sign-off.

Metric	Result
Total Test Cases	9
Test Cases Executed	9
Passed - Initial Execution	7
Failed - Initial Execution	2
Blocked	0
Initial Pass Percentage	77.78%
Defects Identified	2
Defects Resolved	2
Defects Retested	2
Final Passed After Retesting	9
Final Pass Percentage	100%

Defect Summary:

Two medium-severity issues were identified during initial Milestone 4 testing. Both affected dashboard usability or final integration behavior but did not block the complete application. The issues were corrected and verified through retesting before deployment.

Defect ID	Module	Issue	Severity	Status
BUG-05	Dashboard	Pipeline value card did not refresh immediately after applying a dashboard filter.	Medium	Resolved
BUG-06	Follow-up Panel	Follow-up recommendation panel did not update after the related lead stage changed.	Medium	Resolved

Severity

- Critical: Major functionality is completely blocked.
- High: Important functionality is seriously affected.
- Medium: A specific feature is affected but the application can still be used.
- Low: Minor UI or usability issue.

Both defects were classified as Medium because they affected the final dashboard workflow but did not prevent the platform from being used.

Defect Resolution & Retesting:

The identified issues were discussed with the relevant team members and corrected. The affected dashboard and follow-up scenarios were then executed again to confirm that the updated data was displayed correctly.

Defect ID	Issue	Retest Result	Final Status
BUG-05	Dashboard pipeline value refresh after filter	Passed	Closed
BUG-06	Follow-up panel refresh after lead-stage update	Passed	Closed

Summary

Both identified defects were fixed and successfully verified through retesting. After the fixes, all documented Milestone 4 test scenarios passed successfully, allowing the application to proceed to deployment and project closure.

Integration Testing:

Integration testing was completed to make sure that all six SalesGenie AI modules were working together correctly. The complete flow checked was:

Frontend -> FastAPI Backend -> Database / AI / CRM / Analytics Services -> Backend -> Frontend

Areas Checked

- Whether lead, outreach, scoring, CRM, and analytics information moved through the integrated workflow.
- Whether dashboard metrics reflected the information returned by the analytics API.
- Whether pipeline and follow-up recommendations updated correctly after final integration.
- Whether earlier modules continued to work during full regression testing.
- Whether the complete application could be used from lead handling through dashboard review.

Dashboard / Performance / UAT Validation:

Milestone 4 validation focused on final usability, end-to-end stability, and readiness for deployment.

Dashboard and Analytics Validation

- Analytics API values for conversion rate, pipeline value, and KPIs were reviewed in the dashboard.
- Dashboard cards and pipeline view were checked for correct display.
- The pipeline-value refresh issue was corrected and retested successfully.

Performance and Integration Validation

- End-to-end integration of all six modules was completed.
- Performance tuning was reviewed against the <5 second response target.
- Full regression testing was completed before final sign-off.

UAT and Deployment Validation

- UAT with sample users was completed and reviewed against the >=85% satisfaction target.
- The deployed build, final demo flow, and documentation handover were completed.
- The follow-up panel refresh issue was corrected and verified before closure.

QA Observation

The final QA review confirmed that the application workflow was stable across its connected modules and that the dashboard and follow-up features were ready for final use. Functional, regression, integration, UAT, and deployment validation were included in the Day 29 and Day 30 closure activities.

Overall QA Results:

Testing Area	Result
Analytics API Validation	Passed
Dashboard UI and KPI Display	Passed
Kanban Pipeline View	Passed
Follow-up Recommendation Panel	Passed
Full Integration Across Six Modules	Passed
Performance Review	Passed
Full Regression Testing	Passed
UAT with Sample Users	Passed
Defect Retesting	Passed
Deployment Validation	Passed
Milestone 4 QA Review / Sign-off	Completed and Signed Off

Overall QA Summary

Milestone 4 testing covered the Dashboard & Sales Analytics module together with full integration, performance review, regression testing, UAT, deployment validation, and project closure. The review included analytics API checks, dashboard and pipeline validation, follow-up recommendation validation, and end-to-end testing across all six modules.

Two medium-severity issues were identified during initial execution. Both were resolved and successfully retested, and all documented final-milestone scenarios passed in the final execution.

Based on the completed validation, UAT review, deployment check, and final demo, Milestone 4 was approved and signed off.

Conclusion:

Milestone 4 QA was completed to ensure that the Dashboard & Sales Analytics features and the fully integrated SalesGenie AI platform were working properly before deployment. The QA activities covered analytics, dashboard UI, pipeline view, follow-up recommendations, full regression testing, integration testing, performance review, UAT, and deployment validation.

The issues identified during testing were corrected and verified through retesting. Overall, the application was tested, reviewed, deployed, and signed off successfully, completing the final milestone and project closure activities.

Milestone Sign-off:

Milestone	QA / Review Date	Validation Coverage	Final Status
Milestone 4 - Dashboard & Analytics + Full Integration	24-Aug-2026 to 25-Aug-2026	Analytics, dashboard, integration, regression, performance, UAT, defect retesting, and deployment	Completed and Signed Off